Type MS · Area between two curves (or a curve and a line)

Answer 1 of 11 · Question 5 part c · 7 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks
5(a)	$y = Q(x-6)(x+2)$ where Q is a constant	M1
	Using the coordinates $(4,-6)$	
	$-6 = Q(4-6)(4+2) \Rightarrow Q = \frac{-6}{-12} = \frac{1}{2}$	M1
	$y = \frac{1}{2}(x-6)(x+2) \Rightarrow y = \frac{x^2}{2} - 2x - 6$ *	A1 cso
		[3]
(b)	$\frac{dy}{dx} = \frac{2x}{2} - 2$	M1
	$x = 4, \quad \frac{dy}{dx} = 4 - 2 = 2$	M1
	Gradient of normal is $-\frac{1}{2}$	B1ft
	Equation of l: $y - (-6) = -\frac{1}{2}(x-4)$	M1
	$\Rightarrow y = -\frac{x}{2} - 4 \Rightarrow 2y + x + 8 = 0$ *	A1 cso
		[5]
(c)	$\frac{x^2}{2} - 2x - 6 = -\frac{x}{2} - 4 \Rightarrow \frac{x^2}{2} - \frac{3}{2}x - 2 = 0$	M1
	$\Rightarrow (x-4)(x+1) = 0 \Rightarrow x = -1, 4$	M1 A1
	Area $= \int_{-1}^{4}\left(\frac{x^2}{2} - 2x - 6\right) dx - \int_{-1}^{4}\left(-\frac{x}{2} - 4\right) dx$	M1
	$= \left[\frac{x^3}{6} - \frac{3}{4}x^2 - 2x\right]_{-1}^{4}$	M1
	$= \left(\frac{4^3}{6} - \frac{3}{4}\times 4^2 - 2\times 4\right) - \left(\frac{(-1)^3}{6} - \frac{3}{4}\times(-1)^2 - 2\times(-1)\right)$	M1
	$= \left(\frac{4^3}{6} - \frac{3}{4}\times 4^2 - 2\times 4\right) - \left(\frac{(-1)^3}{6} - \frac{3}{4}\times(-1)^2 - 2\times(-1)\right)$	
	$= -\frac{125}{12} \Rightarrow$ Area $= \frac{125}{12}$ (units^2) oe	A1
		[7]
		Total 15 marks

Part	Mark	Notes
(a)	M1	Uses the intersections with the x-axes to form a quadratic equation of the form $y = Q(x \pm 6)(x \pm 2)$

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	M1	Uses their Quadratic with the coordinates $(4, -6)$ to find the value of Q Allow just one processing error here.
	A1 cso	For the correct equation in the required form. Note this equation is given to candidates. Both above steps must be complete and correct for the award of this mark.
	ALT – Uses simultaneous equations	
	M1	Sets up all three equations with the given coordinates. These must be correct. $y = px^2 + qx + r$ $0 = 4p - 2q + r$ **1** $0 = 36p + 6q + r$ **2** $-6 = 16p + 4q + r$ **3**
	M1	Attempts to solve their three simultaneous equations to find the values of p, q and r At least one correct value is evidence of correct method. **2 - 1** $0 = 32p + 8q$ **4** **3 - 2** $6 = 20p + 2q$ **5** **5** $\times 4$ $24 = 80p + 8q$ **6** **6 - 4** $24 = 48p \Rightarrow p = \frac{1}{2}$, $24 = 40 + 8q \Rightarrow q = -2$ $0 = 2 + 4 + r \Rightarrow r = -6$
	A1 cso	For the correct equation in the required form. Note this equation is given to candidates. All of the above steps must be complete and correct for the award of this mark.
(b)	M1	For differentiating the given expression. This must be correct, simplified or unsimplified for this mark.
	M1	For substituting $x = 4$ into their $\frac{dy}{dx}$ to find the gradient of the tangent.
	B1ft	For finding the gradient of the normal. Ft their gradient.
	M1	For forming an equation for l using the equation of the normal which must have come from use of calculus. If they use $y = mx + c$ then they must find c and form an equation for the award of this mark. For example; $c = -4 \Rightarrow y = -\frac{x}{2} - 4$
	A1 cso	For the correct equation in the required form. Accept the terms in any order. For example: even $0 = -8 - x - 2y$
(c)	M1	For equating the equation of S to their l and forming a 3TQ
	M1	For attempting to solve the 3TQ [see General Guidance for the definition of an attempt] to find two points of intersection.
	A1	For **both** correct values of x
	M1	For a correct statement for the area using **their two points of intersection** correctly. **Do not accept limits of $x = -2$ and 4 or 6** They may complete these two areas separately and combine at the end. Check to the end of their work before you score this mark. Accept either $\int_{-1}^{4} \text{Curve} - \int_{-1}^{4} \text{Line}$ or $\int_{-1}^{4} \text{Line} - \int_{-1}^{4} \text{Curve}$ for this mark.
	M1	For an attempt to integrate either the expression for the line or the curve.

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		See General guidance – but no power of x is to decrease.
	M1	For substituting in their values correctly **AND** subtract the two integrals. Explicit substitution must be seen if the final area is incorrect, or the limits are incorrect. A final correct area which follows correct integration is adequate evidence.
	A1	For the correct area

Question	Scheme	Marks

Type MS · Area between two curves (or a curve and a line)

Answer 2 of 11 · Question 10 part d · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

Number		
10 (a)	$6x-x^2=-\left(x^2-6x\right)$ $-\left(x^2-6x\right)=-\left\{(x-3)^2-9\right\}\Rightarrow \mathrm{f}(x)=-(x-3)^2+9$ $D=-1,\ E=-3$ and $F=9$	M1A1A1 [3]
(b)	(i) $\mathrm{f}(x)_{\max}=9$ (ii) $x=3$	B1ft B1ft [2]
(c)	$6x-x^2=x^2-4x+8\Rightarrow 2x^2-10x+8=0$ $2x^2-10x+8=(2x-2)(x-4)\Rightarrow x=1,\ x=4$ $y=5,\ y=8$ Coordinates are (1, 5) and (4, 8)	M1 M1A1 A1 [4]
(d)	$\text{Area}=\int_1^4\left(6x-x^2\right)\mathrm{d}x-\int_1^4\left(x^2-4x+8\right)\mathrm{d}x=\int_1^4\left[-2x^2+10x-8\right]\mathrm{d}x$ $=\left[\frac{-2x^3}{3}+\frac{10x^2}{2}-8x\right]_1^4$ $=\left(\frac{-2\times4^3}{3}+\frac{10\times4^2}{2}-8\times4\right)-\left(\frac{-2\times1^3}{3}+\frac{10\times1^2}{2}-8\times1\right)=9\ (\text{units}^2)$	M1 M1 M1A1 [4]
		Total 13 marks

Type MS · Area between two curves (or a curve and a line)

Answer 2 of 11 · Question 10 part d · 4 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Area between two curves (or a curve and a line)

Answer 3 of 11 · Question 7 part d · 5 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

7(a)	$x^2-9x+14=\left(x-\frac{9}{2}\right)^2+14-\frac{81}{4}=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$	M1
	$a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe	A1 (2)
(b)	(i) least value of $f(x)=-\frac{25}{4}$	B1ft
	(ii) least value when $x=\frac{9}{2}$	B1ft (2)
(c)	$x+5=x^2-9x+14$	M1
	$x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$	M1
	Points are $(9,14)$ $(1,6)$	A1A1 (4)
(d)	Area $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)dx=\int_1^9\left(-x^2+10x-9\right)dx$	M1
	$=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$	M1A1
	$=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=85\frac{1}{3}$	M1A1 (5)
		[13]

Part	Mark	Notes
(a)	M1	For attempting to complete the square to achieve as a minimum $x^2-9x+14=\left(x\pm\frac{9}{2}\right)^2+14-k$ where k is a constant
	A1	For the correct expression $x^2-9x+14=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$ or $a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe stated
(b)(i)	B1ft	For the correct value $f(x)=-\frac{25}{4}$ follow through their value of '$-\frac{25}{4}$'
(ii)	B1ft	For the correct value of $x=\frac{9}{2}$ provided they have $\left(x-\frac{9}{2}\right)^2$ …in part (a). Follow through their value of '$\frac{9}{2}$'

Type MS · Area between two curves (or a curve and a line)

Answer 3 of 11 · Question 7 part d · 5 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

(c)	M1	For equating the equation of the line with the equation of C $x+5=x^2-9x+14\Rightarrow x^2-10x+9=0$ **and** attempting to form a 3TQ
	M1	Attempts to solve their 3TQ by any method, provided it is the result of equating the line with C $x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$
	A1	For the correct coordinates of **either** $(9,14)$ or $(1,6)$
	A1	For both correct pairs of coordinates $(9,14)$ and $(1,6)$
(d)	M1	For a correct expression for the required area with both limits correct. (ft their limits from (c)) Award this mark if they have 'curve – line' but otherwise correct. $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)\mathrm{d}x=\left[\int_1^9\left(-x^2+10x-9\right)\mathrm{d}x\right]$, accept $\int_1^9\left(x^2-10x+9\right)\mathrm{d}x$ **OR** Area under the trapezium – curve $\frac{1}{2}\times8\times(6+14)-\int_1^9\left(x^2-9x+14\right)\mathrm{d}x$
	M1	For attempting to integrate the equation for the combined expression or the curve only.
	A1	For the correct integrated expression for required area. Ignore limits for this mark – even if they are absent altogether. Area = $=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$ accept $\left[\frac{x^3}{3}-5x^2+9x\right]_1^9$ **OR** $\frac{1}{2}\times8\times(6+14)-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ **OR** $\left(\frac{x^2}{2}+5x\right)_1^9-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ or $\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9-\left(\frac{x^2}{2}+5x\right)_1^9$
	M1	For substituting their limits (x – coordinates from part(c)) into their integrated expression. $=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=\ldots$ **OR** $80-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$ **OR** $\left[\left(\frac{9^2}{2}+5\times9\right)-\left(\frac{1^2}{2}+5\times1\right)\right]-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$
	A1	For the correct area of $85\frac{1}{3}$ or $\frac{256}{3}$ If they get a value of $-85\frac{1}{3}$ they must give a **final** value of $85\frac{1}{3}$ for this mark.

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Question Number	Scheme	Marks
	e^x	

Type MS · Area between two curves (or a curve and a line)

Answer 4 of 11 · Question 10 part b · 9 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

10 (a)	$\sin x + 1 = \cos x + 1 \Rightarrow \tan x = 1$	M1
	$x = \frac{\pi}{4}, \frac{5\pi}{4}$	A1 A1 (3)
(b)	Area R_1 :	
	$\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x + 1) - (\cos x + 1)\,dx = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x)\,dx$	M1
	$\left[-\cos x - \sin x\right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$	M1
	$\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) - \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right) = 2\sqrt{2}$	M1 A1
	Area R_2 :	
	$\int_{\pi}^{\frac{5\pi}{4}} (\cos x + 1)\,dx + \int_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} (\sin x + 1)\,dx$	M1
	$\left[\sin x + x\right]_{\pi}^{\frac{5\pi}{4}} + \left[-\cos x + x\right]_{\frac{5\pi}{4}}^{\frac{3\pi}{2}}$	A1
	$\left[\left(-\frac{\sqrt{2}}{2} + \frac{5\pi}{4}\right) - \pi\right] + \left[\frac{3\pi}{2} - \left(\frac{\sqrt{2}}{2} + \frac{5\pi}{4}\right)\right] = -\sqrt{2} + \frac{1}{2}\pi$	M1 A1
	area of R_1 : area of $R_2 = 2 : \left(\frac{\pi\sqrt{2}}{4} - 1\right)$ oe	A1 (9)
		Total 12 marks

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Answer 4 of 11 · Question 10 part b · 9 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For $\tan x = 1$
	A1	For $x = \frac{\pi}{4}$ [Allow 45°]
	A1	For $x = \frac{5\pi}{4}$ [Allow 225°]
(b)	M1	For stating $A = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x + 1) - (\cos x + 1)\,dx$
	M1	For attempting to integrate their expression for the area. This must be correct for this mark. $\int (\sin x - \cos x)\,dx = -\cos x - \sin x$
	M1	For substitution of correct limits
	A1	For $2\sqrt{2}$
	M1	For attempting $\int f(x)dx + \int g(x)dx$ with the correct limits. $\int_{\pi}^{\frac{5\pi}{4}} (\cos x + 1)\,dx + \int_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} (\sin x + 1)\,dx$
	M1	For attempting to integrate, which must be correct for this mark.
	M1	For substitution of their limits correctly.
	A1	For $-\sqrt{2} + \frac{1}{2}\pi$
	A1	For the simplified ratio $2 : \left\{\frac{\pi\sqrt{2}}{4} - 1\right\}$

Type MS · Area between two curves (or a curve and a line)

Answer 4 of 11 · Question 10 part b · 9 marks

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Question	Scheme	Marks

Type MS · Area between two curves (or a curve and a line)

Answer 5 of 11 · Question 11 part c · 8 marks

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Number		
11 (a)	$\cos 2x = \cos(x+x)$	
	$= \cos^2 x - \sin^2 x = 1 - \sin^2 x - \sin^2 x$	M1 M1
	$= 1 - 2\sin^2 x$ *	A1 cso (3)
(b)	$\sin x + 2\sin^2 x - 1 = 0 \Rightarrow (2\sin x - 1)(\sin x + 1) = 0$	M1dM1
	When $\sin x = \frac{1}{2}$ When $\sin x = -1$	
	$x = \frac{\pi}{6}, \frac{5\pi}{6}$ $x = \frac{3\pi}{2}$	A1
	$\left(\frac{\pi}{6}, \frac{5}{2}\right), \left(\frac{5\pi}{6}, \frac{5}{2}\right), \left(\frac{3\pi}{2}, 1\right)$	A1 (4)
(c)	$\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\sin x + 2)\,dx - \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\cos 2x + 2)\,dx$	M1
	$\left[-\cos x + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} - \left[\frac{\sin 2x}{2} + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$	A1
	$\left[\left(-\cos\frac{5\pi}{6} + 2\left(\frac{5\pi}{6}\right)\right) - \left(-\cos\frac{\pi}{6} + 2\left(\frac{\pi}{6}\right)\right)\right]$	
	$-\left[\left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2} + 2\left(\frac{5\pi}{6}\right)\right) - \left(\frac{\sin 2\left(\frac{\pi}{6}\right)}{2} + 2\left(\frac{\pi}{6}\right)\right)\right]$	M1
	$= \frac{3\sqrt{3}}{2}$	A1
	$\int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} (\cos 2x + 2)\,dx - \int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} (\sin x + 2)\,dx$	M1
	$\left[\frac{\sin 2x}{2} + 2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} - \left[-\cos x + 2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}$	
	$\left[\left(\frac{\sin 2\left(\frac{3\pi}{2}\right)}{2} + 2\left(\frac{3\pi}{2}\right)\right) - \left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2} + 2\left(\frac{5\pi}{6}\right)\right)\right]$	M1
	$-\left[\left(-\cos\left(\frac{3\pi}{2}\right) + 2\left(\frac{3\pi}{2}\right)\right) - \left(-\cos\left(\frac{5\pi}{6}\right) + 2\left(\frac{5\pi}{6}\right)\right)\right]$	
	$\frac{3\sqrt{3}}{4}$	A1
	$R_1 : R_2 = 2 : 1$	B1ft
	See Special Case at end of notes	(8)
		Total 15 marks

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Part	Mark	Notes
(a)	M1	For using the addition formula on page 2 and reaching the result $\cos 2x = \cos^2 x - \sin^2 x$ This **must** be correct for this mark.
	M1	For using **and** applying $\sin^2 x + \cos^2 x = 1$ in their $\cos 2x$ $\cos 2x = 1 - \sin^2 x - \sin^2 x$
	A1 cso	For the correct identity as shown in full. $\cos 2x = 1 - 2\sin^2 x$ * **Note: This is a show question, there must be no errors seen.**
(b)	M1	Equates the two given equations together. $\sin x + 2 = \cos 2x + 2$ and uses the given identity in part (a) to attempt to form a 3TQ in sin x: $\sin x + 2\sin^2 x - 1 = 0$ or $2\sin^2 x + \sin x - 1 = 0$ At least either $2\sin^2 x$ or $\sin x$ must be correct.
	dM1	Solves their 3TQ by any method to reach two values for $\sin x$ -see General Guidance $(2\sin x - 1)(\sin x + 1) = 0 \Rightarrow \sin x = \left(-1,\ \frac{1}{2}\right)$ This mark is dependent on the previous M mark.
	A1	For finding all three values of x: $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{3\pi}{2}$ Allow $x = 30^\circ,\ 150^\circ,\ 270^\circ$ for this mark – condone missing degree signs
	A1	For all three sets of correct coordinates: $\left(\frac{\pi}{6}, \frac{5}{2}\right), \left(\frac{5\pi}{6}, \frac{5}{2}\right), \left(\frac{3\pi}{2}, 1\right)$ Allow $\left(30^\circ, \frac{5}{2}\right), \left(150^\circ, \frac{5}{2}\right), \left(270^\circ, 1\right)$ – condone missing degree signs
(c)	**Area R_1**	
	M1	For stating a correct method to find the area R_1 using their limits from part (b) of $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ applied the correct way. Condone angles in degrees. $\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\sin x + 2)\,dx - \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\cos 2x + 2)\,dx$ (The + 2 – 2 may have been simplified).
	M1	For an attempt to integrate their expression for R_1 which must involve $\sin x$ and $\cos 2x$ A minimally acceptable integral of $\sin x \to -\cos x$ **or** $\cos 2x \to \pm\frac{\sin 2x}{2}$ and $2 \to 2x$ (The $2 \to 2x$ may not be present if they've simplified first). At least one trig term must be acceptable **with** $2 \to 2x$ (if present) correct. $\left[-\cos x + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} - \left[\frac{\sin 2x}{2} + 2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$ Ignore incorrect, angles in degrees or even absent limits for this mark.

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	M1	Substitutes their limits correctly into their integrated expression. Must involve a changed expression. $\left[\left(-\cos\frac{5\pi}{6}+2\left(\frac{5\pi}{6}\right)\right)-\left(-\cos\frac{\pi}{6}+2\left(\frac{\pi}{6}\right)\right)\right]-$ $\left[\left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2}+2\left(\frac{5\pi}{6}\right)\right)-\left(\frac{\sin 2\left(\frac{\pi}{6}\right)}{2}+2\left(\frac{\pi}{6}\right)\right)\right]$ **Do not accept a substitution in degrees – it must be in radians.** **If the work for the first two method marks to find the area of R_1 is fully correct, this mark can be implied by $\frac{3\sqrt{3}}{2}$. If the student does not get** $\left[-\cos x+2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}-\left[\frac{\sin 2x}{2}+2x\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$ **from the first two methods marks, they must show the clear substitution of their limits to be awarded this mark.**
	A1	For the correct area.
Area R_2		
	M1	For an attempt to integrate their expression for R_2 which must involve $\int(\cos 2x+2)\,dx-\int(\sin x+2)\,dx$ or $\int\cos 2x-\sin x\,dx$ if they have simplified the +2 – 2. Ignore incorrect, angles in degrees or even absent limits for this mark. A minimally acceptable integral of $\sin x\to-\cos x$ or $\cos 2x\to\pm\frac{\sin 2x}{2}$ and $2\to 2x$ (The $2\to 2x$ may not be present if they've simplified first). At least one trig term must be acceptable **with** $2\to 2x$ (if present) correct. $\left[\frac{\sin 2x}{2}+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}-\left[-\cos x+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}$
	M1	For substituting their limits correctly into their integrated expression. Must involve a changed expression. $\left[\left(\frac{\sin 2\left(\frac{3\pi}{2}\right)}{2}+2\left(\frac{3\pi}{2}\right)\right)-\left(\frac{\sin 2\left(\frac{5\pi}{6}\right)}{2}+2\left(\frac{5\pi}{6}\right)\right)\right]$ $-\left[\left(-\cos\left(\frac{3\pi}{2}\right)+2\left(\frac{3\pi}{2}\right)\right)-\left(-\cos\left(\frac{5\pi}{6}\right)+2\left(\frac{5\pi}{6}\right)\right)\right]$ **Do not accept a substitution in degrees – it must be in radians.** **If the student's work for the first method mark to find the area of R_2 is fully correct, this M mark can be implied by $\frac{3\sqrt{3}}{4}$. If the student does not get**

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		$\left[\frac{\sin 2x}{2}+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}-\left[-\cos x+2x\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}$ **from the first method mark, they must show the clear substitution of their limits to be awarded this mark.**
	A1	For the correct area.
Ratio	**area of R_1: area of R_2**	
	B1ft	For a ratio given in its simplest form $a : b$ where a, b are positive integers, ft their values for the area of R_1 *and* R_2 Correct ratio is $R_1:R_2 = 2 : 1$
	SC	**Special case: For the use of degrees in part (c).** **<u>If the area of R_1 and R_2 are both correct</u> when limits in degrees have been substituted** (possible, for example, when the candidate has simplified before integrating): Penalise the first substitution in degrees for the area R_1 by awarding M1 M1 M0 A0. For the area R_2 allow M1 M1 A1 if the substitution (in degrees) and the area of R_2 is correct. B1ft for the correct ratio.

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Type MS · Area between two curves (or a curve and a line)

Answer 6 of 11 · Question 9 part whole · 8 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

9	**Area under the curve** $A_c = \int_{-1}^{0}\left(-2e^{3x}+4\right)dx, = \left[\frac{-2e^{3x}}{3}+4x\right]_{-1}^{0}$ $\Rightarrow \left(\frac{-2e^{3\times0}}{3}+4\times0\right)-\left(\frac{-2e^{3\times-1}}{3}+4\times-1\right), = \frac{10}{3}+\frac{2e^{-3}}{3}$	M1M1 M1A1
	Area under trapezium Method 1 [uses formula for the area of a trapezium] $y=2 \quad y=-2e^{-3}+4$ $A_T = \frac{1}{2}\times1\times\left('2'+'-2e^{-3}+4'\right)=\left[3-e^{-3}\right]$	B1
	Method 2 [finds equation of the line and integrates] Equation of the line: $y=\left(2e^{-3}-2\right)x+2 \quad$ (accept $y=-1.9x+2$) $A_T = \int_{-1}^{0}'\left(\left(2e^{-3}-2\right)x+2\right)'dx = \left[\frac{'\left(2e^{-3}-2\right)'x^2}{2}+\frac{'2'x}{2}\right]_{-1}^{0} = \left[3-e^{-3}\right]$	M1
	Shaded Area $A_{shaded} = \left(\frac{10}{3}+\frac{2e^{-3}}{3}\right)-\left(3-e^{-3}\right)=\frac{1+5e^{-3}}{3}$	M1A1 [8]
		Total 8 marks

Type MS · Area between two curves (or a curve and a line)

Answer 6 of 11 · Question 9 part whole · 8 marks

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Mark	Notes
M1	For a correct statement for the area under the curve with **correct limits**. This mark can be implied/embedded by later correct work. **Condone missing d*x***
M1	Attempts to integrate the given curve to a form of $Ae^{3x} + Bx$ **Ignore limits for this mark**.
M1	For substituting in both of their limits into their integrated expression in the form $Ae^{3x} + Bx$ and subtracts. (A, B are nonzero constants) Correct exact answer $\frac{10}{3} + \frac{2e^{-3}}{3}$ implies the correct substitution. If their answer is not correct and exact, explicit substitution needs to be seen.
A1	For the correct **exact** area under the curve. If not seen explicitly, this can be implied by correct exact final answer.
Method 1 - trapezium	
B1	For both correct **exact** values of y
M1	For the correct method of finding area of the trapezium using their values for y Look for $\frac{1}{2} \times 1 \times (sum\ of\ their\ y\ values)$
Method 2 – equation of line	
B1	For the correct equation of the line L, must reach $y = \ldots$ $y = \left(2e^{-3} - 2\right)x + 2$ (accept $y = -1.9x + 2$)
M1	For the correct area under the line L using their line. Check for correct integration of their line and full substitution of – 1 if their area is incorrect. (If the terms are equal to zero, substitution of 0 does not need to be seen.) Allow also $A_T = \int_{-1}^{0} '(-1.9x+2)'\,dx = \left[\frac{'-1.9'x^2}{2} + \frac{'2'x}{2}\right]_{-1}^{0}$
Combines the areas	
M1	Correct strategy to find the value of shaded area The integrated area of the curve - the area of trapezium
A1	For the correct **exact** area of the shaded region in the required form
Some candidates are combining the curve – line in one calculation – mark this as follows. • Wherever you see the **correct** equation of the line, score the 5th Mark [B mark] • Mark the work for the curve (which may be embedded in a combined integral) using the first 4 marks. • Mark the area under the line using the 6th mark [M mark – using their equation in the form $y = mx + c$] • When you see them subtracting the areas – award the penultimate M mark If they add the areas, this is M0 • Final A mark is for the correct answer only	

Type MS · Area between two curves (or a curve and a line)

Answer 6 of 11 · Question 9 part whole · 8 marks

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Question	Scheme	Marks

Type MS · Area between two curves (or a curve and a line)

Answer 7 of 11 · Question 9 part b · 7 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

9 a	$(ax^4 = 3x^4) \Rightarrow a = 3$	B1
	$(4c = 64) \Rightarrow c = 16$	B1
	$(bx^3 - 4ax^3 = 4x^3) \Rightarrow b - 4a = 4$ or $(4ax^2 - 4bx^2 + cx^2 = -36x^2)$ $\Rightarrow 4a - 4b + c = -36$ *or* $(4bx - 64x = 0x) \Rightarrow 4b - 64 = 0$	M1
	$b = 16$	A1 (4)
b	$\int_0^2 (x^3 + x^2 - 6x)\mathrm{d}x = \left[\frac{x^4}{4} + \frac{x^3}{3} - \frac{6}{2}x^2\right]_0^2$	B1 (M1 on ePen) M1
	$= \left("\frac{2^4}{4} + \frac{2^3}{3} - 3(2)^2"\right) - (0)$	M1 (A1 on ePen)
	$= \pm\frac{16}{3}$ oe	A1 (M1 on ePen)
	$\left(\left["\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right]_x^0 = \pm\frac{16}{3} \Rightarrow\right) (0) - \left("\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right) = \pm\frac{16}{3}$ oe	M1
	$3x^4 + 4x^3 - 36x^2 + 64 = 0$	A1
	$(x-2)^2(3x^2 + 16x + 16) = 0$ *	A1* cso (7)
ALT	$\int_x^2 (x^3 + x^2 - 6x)\mathrm{d}x = \left["\frac{x^4}{4} + \frac{x^3}{3} - 3x^2"\right]_x^2$	B1 (M1 on ePen) M1
	$= \left("\frac{2^4}{4} + \frac{2^3}{3} - 3(2)^2"\right) - \left("\frac{x^4}{4} + \frac{x^3}{3} - 3(x)^2"\right)$	M1 (A1 on ePen)
	$\pm\left(-\frac{16}{3} - \frac{x^4}{4} - \frac{x^3}{3} + 3x^2\right) = 0$	A1 (M1 on ePen)
	$3x^4 + 4x^3 - 36x^2 + 64 = 0$	M1 A1
	$(x-2)^2(3x^2 + 16x + 16) = 0$ *	A1
c	(When $(x-2) = 0$ $x = 2$ and when $3x^2 + 16x + 16 = 0$) $(3x+4)(x+4) = 0$ so $x = -\frac{4}{3}$ and $x = -4$	M1 A1
	($x \neq -4$ as it is to the left of the point at $x = -3$) $x = -\frac{4}{3}$	M1
	When "$x = -\frac{4}{3}$" $y = \left("-\frac{4}{3}"\right)^3 + \left("-\frac{4}{3}"\right)^2 - 6\left("-\frac{4}{3}"\right) = \frac{200}{27}$	dM1
	So $A = \left(-\frac{4}{3}, \frac{200}{27}\right)$	A1 (5)
		Total 16 marks

Type MS · Area between two curves (or a curve and a line)

Answer 7 of 11 · Question 9 part b · 7 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	$a = 3$
	B1	$c = 16$
	M1	For a clear process to equate coefficients to find either of the equations $b-4a=4$ or $4a-4b+c=-36$ or $4b-64=0$. Allow one error. For algebraic division: there must be a complete attempt to divide by x^2+px+q and reach an expression of the form $3x^2+rx+16,\ p,q,r\neq 0$
	A1	$b=16$
Full marks may be awarded for all of a, b and c appearing correctly with no method shown. Embedded a, b and c should be awarded full marks.		
(b)	B1 (M1 on ePen)	Correct limits of $x = 0$ and $x = 2$ – or used later in working.
	M1	For a minimally acceptable attempt to integrate any 3 term cubic of the form x^3+fx^2+gx . Limits do not need to be present. See general guidance for the definition of a minimally acceptable attempt.
	M1 (A1 on ePen)	Substitution of their limits into any changed expression, the correct way round minimum 3 terms. Sub of 0 not needed. Allow this mark to be implied by a fully correct value.
	A1 (M1 on ePen)	Either for a fully correct substitution, or for $\pm\frac{16}{3}$
	M1	For $(0)-\left("\frac{x^4}{4}+\frac{x^3}{3}-3x^2"\right)=\pm\frac{16}{3}$ oe. Allow use of their integrated expression and their $\pm\frac{16}{3}$ which can be in any equivalent form, including an unevaluated expression. Allow use of n rather than x. The limit of 0 need not be seen.
	A1	For $3x^4+4x^3-36x^2+64=0$ Allow use of n rather than x. $\frac{16}{3}$ must have been evaluated.
	A1* cso	For $(x-2)^2(3x^2+16x+16)=0$ .No errors in working. Candidates can use n throughout.
ALT	B1 (M1 on ePen)	Correct limits of $x = x$ and $x = 2$ – seen on the integral or used later in working. These can be either way round.
	M1	For a minimally acceptable attempt to integrate any 3 term cubic of the form x^3+fx^2+gx . Limits do not need to be present. See general guidance for the definition of a minimally acceptable attempt.
	M1 (A1 on ePen)	For substitution of both of their limits into any changed expression. Minimum 3 terms
	A1 (M1 on ePen)	For $\pm\left(-\frac{16}{3}-\frac{x^4}{4}-\frac{x^3}{3}+3x^2\right)=0$
	M1	For a valid attempt to multiply their equation throughout to arrive at integer coefficients.
	A1 A1	As main scheme. Candidates can use n throughout.
(c)	M1	For a complete and minimally acceptable attempt to solve $3x^2+16x+16=0$ to give $x =$ See general guidance for definition of minimally acceptable. Some candidates may have factorised in part b), if used in part c), marks can be awarded
	A1	For $x=-\frac{4}{3}$ and $x=-4$
	M1	Chooses $"x=-\frac{4}{3}"$, can be stated explicitly or used later implicitly. Allow choice of their x coordinate for $-3<x<0$
	dM1	For substituting $x="-\frac{4}{3}"$ into $y=x(x+3)(x-2)$ Allow substitution of their x, dependent on previous method mark
	A1	$A=\left(-\frac{4}{3},\frac{200}{27}\right)$ students may list $x=-\frac{4}{3}, y=\frac{200}{27}$

Type MS · Area between two curves (or a curve and a line)

Answer 7 of 11 · Question 9 part b · 7 marks

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Question number	Scheme	Marks
	→ → 3 → 3 → 1 → 1	

Type MS · Area between two curves (or a curve and a line)

Answer 8 of 11 · Question 11 part d · 5 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

Question	Scheme	Marks
11(a)	$f(x)=10+6x-x^2=-(x^2-6x)+10 \Rightarrow A=-1$	B1
	$f(x)=-\left[(x-3)^2-9\right]+10=-(x-3)^2+19$	M1
	$B=-3,\ C=19$	A1A1
		[4]
(b)	(i) $x=3$	B1ft
	(ii) f $(x)_{\text{greatest}} = 19$	B1ft
		[2]
(c)	$x^2-x+13=10+6x-x^2 \Rightarrow 2x^2-7x+3=0$	M1
	$\Rightarrow (2x-1)(x-3)=0 \Rightarrow x=\frac{1}{2},\ 3$	M1A1
		[3]
(d)	$A=\int_{\frac{1}{2}}^{3}(10+6x-x^2)\,dx-\int_{\frac{1}{2}}^{3}(x^2-x+13)\,dx=\left[\int_{\frac{1}{2}}^{3}-2x^2+7x-3\,dx\right]$	M1
	$A=\left[-\frac{2x^3}{3}+\frac{7x^2}{2}-3x\right]_{\frac{1}{2}}^{3}$	M1A1
	$A=\left[\left(-\frac{2\times3^3}{3}+\frac{7\times3^2}{2}-3\times3\right)-\left(-\frac{2\times0.5^3}{3}+\frac{7\times0.5^2}{2}-3\times0.5\right)\right]$	M1
	$A=\frac{125}{24}$ oe	A1
		[5]
		Total 14 marks

Part	Mark	Notes
(a)	**B1**	For factorising – 1 and finding $A = -1$
	M1	For an attempt to complete the square – (See general guidance)
	A1	For either $B=-3$ or $C=19$
	A1	For both $B=-3$ and $C=19$ Accept all values embedded.
	NB: Correct values following no working – Award full marks in this part.	

Type MS · Area between two curves (or a curve and a line)

Answer 8 of 11 · Question 11 part d · 5 marks

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(b)(i)	**B1ft**	$x = 3$ ft their B You may see differentiation to find a maximum x. Allow a correct value of x here even if it does not follow from their working.
(b)(ii)	**B1ft**	f $(x) = 19$ ft their C You may see differentiation to find a maximum x. Allow a correct value of f(x) here even if it does not follow from their working. Allow $y = 19$.
(c)	**M1**	Sets the equation of curve C equal to the equation of curve S and forms a 3TQ
	M1	Attempts to solve their 3TQ using any correct method to find **two** values of x If their 3TQ is incorrect and no working is seen to solve it, withhold this mark.
	A1	For both $x = \frac{1}{2}$, 3 correct Accept coordinates – ignore y values.
	NB: Both correct values following no working – Award full marks in this part.	
(d)	**M1**	For a correct statement with their values of x in the correct position of the intent to integrate the two expressions and **subtract** the result, or to integrate $2x^2 - 7x + 3$ Ft their limits for this mark, the correct way around, but allow recovery at a later stage. If they deal with the two expressions separately, look for subtraction at the end of their solution.
	M1	For an attempt to integrate their combined expression or **two** separate expressions. They must achieve as a minimum $\pm Px^3 \pm Qx^2 \pm Rx$ for their integration whether they integrate two expressions separately (each) or one combined expression. Ignore limits for this mark.
	A1	For the correct integrated expression(s)– ignore limits for this mark Allow $\frac{2x^3}{3} - \frac{7x^2}{2} + 3x$
	M1	For substituting their limits into their integrated expression and subtracting the results of lower limit from upper limit. **Note carefully:** If they have the correct limits and correct integration allow this mark if the **correct** area is obtained without seeing explicit substitution. $\left(\pm\frac{125}{24},\ \pm 5.208...\right)$ If their limits are incorrect or the integration is incorrect or the final area is incorrect then explicit substitution **must** be seen for the award of this mark.
	A1	For the correct area $A = \frac{125}{24}$ oe

Type MS · Area between two curves (or a curve and a line)

Answer 8 of 11 · Question 11 part d · 5 marks

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Type MS · Area between two curves (or a curve and a line)

Answer 9 of 11 · Question 10 part b · 6 marks

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10 (a)	$\int\left(3x^2-4x-p\right)\,\mathrm{d}x=\frac{3x^3}{3}-\frac{4x^2}{2}-px+c\quad\left[=x^3-2x^2-px+c\right]$	M1A1
	$y=x^3-2x^2-px+c$ At $(2,0)\quad 0=8-8-2p+c\Rightarrow c=2p$ At $(-1,9)\quad 9=-1-2+p+c\Rightarrow c=12-p$ $\Rightarrow p=4,c=8\Rightarrow y=x^3-2x^2-4x+8$*	M1 M1 A1A1cso [6]
(b)	$x^3-2x^2-4x+8=8-4x\Rightarrow x^3-2x^2=0\Rightarrow x^2\left(x-2\right)=0$ $x=0,x=2$	M1 A1
	Area $=\int_0^2\left(8-4x\right)\,\mathrm{d}x-\int_0^2\left(x^3-2x^2-4x+8\right)\,\mathrm{d}x$ Area $=\int_0^2\left(8-4x\right)\,\mathrm{d}x-\int_0^2\left(x^3-2x^2-4x+8\right)\,\mathrm{d}x=\int_0^2\left(-x^3+2x^2\right)\,\mathrm{d}x$ Area $=\left[-\frac{x^4}{4}+\frac{2x^3}{3}\right]_0^2=\left(-4+\frac{16}{3}\right)-\left(0\right)=\frac{4}{3}$	M1 M1M1A1 [6]
		Total 12 marks

Type MS · Area between two curves (or a curve and a line)

Answer 9 of 11 · Question 10 part b · 6 marks

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Question number	Scheme	Marks

Type MS · Area between two curves (or a curve and a line)

Answer 10 of 11 · Question 8 part b · 6 marks

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8 (a)(i)	$e^{3x} - 1 = 9 - 9e^{-3x}$	M1
	$(e^{3x})^2 - e^{3x} = 9e^{3x} - 9$	M1
	$\left(e^{3x}\right)^2 - 10e^{3x} + 9 = 0$ *	A1cso*
(ii)	$(e^{3x} - 1)(e^{3x} - 9)$ [= 0] leading to $e^{3x} =$	M1
	$e^{3x} = 9 \quad \rightarrow \quad x = \frac{1}{3}\ln 9$ *	A1cso* [5]
Note: subscripts on marks in (b) indicate which mark is being awarded on epen (1=1st etc)		
(b)	$\left(\int_0^{\frac{1}{3}\ln 9}(9 - 9e^{-3x})\,dx =\right)$ $9x + 3e^{-3x}$ and $\left(\int_0^{\frac{1}{3}\ln 9}(e^{3x} - 1)\,dx =\right)$ $\frac{e^{3x}}{3} - x$	$M1_1$ $A1_2A1_3$
	$"9x + 3e^{-3x}" - "\frac{e^{3x}}{3} - x"$ $M1_4$ $\left(10 \times \frac{1}{3}\ln 9 - \frac{9}{-3}e^{-3\times\frac{1}{3}\ln 9} - \frac{e^{3\times\frac{1}{3}\ln 9}}{3}\right) - \left(10 \times 0 - \frac{9}{-3}e^{-3\times 0} - \frac{e^{3\times 0}}{3}\right)$ $dM1_5$	$\left(9 \times \frac{1}{3}\ln 9 + 3e^{-3\times\frac{1}{3}\ln 9}\right) - (9 \times 0 + 3e^0)$ $dM1_5$ and $\left(\frac{e^{3\times\frac{1}{3}\ln 9}}{3} - \frac{1}{3}\ln 9\right) - \left(\frac{e^0}{3} - 0\right)$ $"3\ln 9 - \frac{8}{3}" - \left("\frac{8}{3} - \frac{1}{3}\ln 9"\right)$ $M1_4$
	$\frac{10}{3}\ln 9 - \frac{16}{3}$	$A1_6$
	ALT $\int(9 - 9e^{-3x}) - (e^{3x} - 1)dx$	$M1_4$
	$(\int(10 - 9e^{-3x} - e^{3x})\,dx =)\ 10x - \frac{9}{-3}e^{-3x} - \frac{e^{3x}}{3}$	$M1_1$ $A1_2$ $A1_3$
	$\left(10\times\frac{1}{3}\ln 9 - \frac{9}{-3}e^{-3\times\frac{1}{3}\ln 9} - \frac{e^{3\times\frac{1}{3}\ln 9}}{3}\right) - \left(10\times 0 - \frac{9}{-3}e^{-3\times 0} - \frac{e^{3\times 0}}{3}\right)$	$dM1_5$
	$\frac{10}{3}\ln 9 - \frac{16}{3}$ oe	$A1_6$ [6]
		Total 11 marks

Type MS · Area between two curves (or a curve and a line)

Answer 10 of 11 · Question 8 part b · 6 marks

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Part	Mark	Additional Guidance
Mark parts (i) and (ii) together.		
(a)(i)	M1	For equating the two equations.
	M1	For multiplying through by e^{3x}, minimum of 2 out of 4 correct terms. (presence of $\pm 10e^{3x}$ indicates 2 correct terms).
	A1*cso	Correct solution only, no errors or omissions.
(ii)	M1	Minimally acceptable attempt at solving the equation leading to $e^{3x} =$ See general guidance, if the formula is quoted allow up to two slips in substitution, otherwise the substitution must be correct.
	A1*cso	Correct solution only, no errors or omissions. If 0 also included then this should be rejected.
(b)	$M1_1$	For attempt to integrate one of: $9 - 9e^{-3x}$ or $e^{3x} - 1$ or $\pm[(9 - 9e^{-3x}) - (e^{3x} - 1)]$ Limits may not be present. At least one term correct. Ignore +c if included.
	$A1_2$	For correct integration of one of the exponential terms $\pm 9e^{-3x} \rightarrow \mp\frac{9}{3}e^{-3x}$ **or** $\pm e^{3x} \rightarrow \pm\frac{1}{3}e^{3x}$ Limits need not be present. Ignore +c if included.
	$A1_3$	For correct integration of both curves $9x + 3e^{-3x}$ **and** $\frac{e^{3x}}{3} - x$ or for a **fully correct** integration where the difference between two expressions is found $\pm\left(10x - \frac{9}{-3}e^{-3x} - \frac{e^{3x}}{3}\right)$ or $\pm\left(9x - \frac{9}{-3}e^{-3x} - \frac{e^{3x}}{3} + x\right)$ **Note: this is an M mark in epen**
	$M1_4$	For the difference between the two expressions either before or after integration. Allow subtraction either way around. **Note: this is an A mark in epen**
	$dM1_5$	Substitution of correct limits into their integrated expressions (limits subtracted the correct way around). Dependent on first M scored. If substituting before difference found then must substitute into both integrated expressions. May be implied by awrt 1.99 If integration is not correct then substitution must be shown.
	$A1_6$	For the correct answer oe. Must be exact value.

Type MS · Area between two curves (or a curve and a line)

Answer 10 of 11 · Question 8 part b · 6 marks

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Question number	Scheme	Marks

Type MS · Area between two curves (or a curve and a line)

Answer 11 of 11 · Question 8 part c · 5 marks

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8(a)	$\int 17+2x-3x^2\ \mathrm{d}x = 17x+\frac{2x^2}{2}-\frac{3x^3}{3}+k$	M1A1
	$0=\left(17\times(-1)+\frac{2\times(-1)^2}{2}-\frac{3\times(-1)^3}{3}\right)+k \Rightarrow k=15$	M1
	$y=15+17x+x^2-x^3$	A1 [4]
(b)	$\frac{(15+17x+x^2-x^3)}{(x+1)}=-x^2+2x+15$	M1A1
	$-x^2+2x+15=(x+3)(5-x)$	M1A1
	$a=-3,\ [-1]$ and $b=5$	A1
	When $x=0$ $y=15$ so $c=15$	B1 [6]
(c)	$\int_0^5(15+17x+x^2-x^3)\ \mathrm{d}x-\frac{1}{2}\times5\times15$	M1
	OR	
	$\int_0^5(15+17x+x^2-x^3)\ \mathrm{d}x-\int_0^5(15-3x)\ \mathrm{d}x$	
	$\int_0^5(15+17x+x^2-x^3)\ \mathrm{d}x=\left[15x+\frac{17x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\right]_0^5$	M1
	$\left(15\times5+\frac{17\times5^2}{2}+\frac{5^3}{3}-\frac{5^4}{4}\right)-(0)=\left(\frac{2075}{12}\right)$	M1
	Area of triangle Area $=\frac{1}{2}\times5\times15=37.5$ **OR**	B1
	Area $=\int_0^5(15-3x)\ \mathrm{d}x=\left[15x-\frac{3x^2}{2}\right]_0^5=37.5$	[B1]
	For the correct area of $R=\frac{2075}{12}-37\frac{1}{2}=\frac{1625}{12}=135\frac{5}{12}$	A1 [5]
		Total 15 marks

Type MS · Area between two curves (or a curve and a line)

Answer 11 of 11 · Question 8 part c · 5 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 4 · find part c in this mark scheme extract

Question	Notes	Marks
8(a)	$f'(x) = 17 + 2x - 3x^2$	
	For an attempt to integrate $f'(x)$ $y = 17x + \frac{2x^2}{2} - \frac{3x^3}{3} + (k)$	M1
	For the correct integral **including** a constant of integration $y = 17x + \frac{2x^2}{2} - \frac{3x^3}{3} + k$	A1
	For substituting $(-1\ ,\ 0)$ into their integrated expression, which must include a constant of integration. $0 = \left(17\times(-1) + \frac{2\times(-1)^2}{2} - \frac{3\times(-1)^3}{3}\right) + k \Rightarrow (k = 15)$	M1
	For writing the equation in the required form $y = 15 + 17x + x^2 - x^3 *$ **This is a given equation. Every step above must be seen for the award of full marks.**	A1 cso [4]
(b)	Divides $(15 + 17x + x^2 - x^3)$ by $(x+1)$ $x+1 \overline{)15 + 17x + x^2 - x^3}$ $\quad Q + 2x + x^2$ **OR** Equates coefficients $(15 + 17x + x^2 - x^3) = (x+1)(Ax^2 + Bx + c) \Rightarrow (x+1)(-x^2 + 2x + Q)$ Where Q is a constant **Minimal working here is sight of the quadratic factor.**	M1
	For obtaining the correct 3TQ $-x^2 + 2x + 15$	A1
	For factorising their 3TQ [or otherwise solving] $-x^2 + 2x + 15 = (x+3)(5-x)$	M1A1
	For $a = -3$, $[-1]$ and $b = 5$ identified clearly.	A1
	SC No working seen [use of a root finder on a calculator]	
	For $(x+1)(x-5)(x+3)$ seen leading to $a = -3$ and $b = 5$ with no working award: M0A0MA1A1	
	For $a = -3$ and $b = 5$ seen with no other working seen award: M0A0M0A0A1	
	For the value of $c = 15$	B1
(c)	For writing a correct expression for the area of R with the correct limits $\int_0^5 (15 + 17x + x^2 - x^3)\ dx - \frac{1}{2}\times 5 \times 15$ OR	M1

Type MS · Area between two curves (or a curve and a line)

Answer 11 of 11 · Question 8 part c · 5 marks

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	$\int_0^5 \left(15+17x+x^2-x^3\right)\,\mathrm{d}x - \int_0^5 (15-3x)\,\mathrm{d}x$	
	For an attempt to integrate the expression for the curve. [Ignore limits for this mark] Area $= \int_0^5 \left(15+17x+x^2-x^3\right)\,\mathrm{d}x = \left[15x+\frac{17x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\right]_0^5$	M1
	For evaluating their integral using their limits. Area $= \left(15\times5+\frac{17\times5^2}{2}+\frac{5^3}{3}-\frac{5^4}{4}\right)-(0)=\left(\frac{2075}{12}\right)$	M1
	For the area of the triangle Area $=\frac{1}{2}\times5\times15=37.5$ **ALT** Integrates the line Area $= \int_0^5 (15-3x)\,\mathrm{d}x = \left[15x-\frac{3x^2}{2}\right]_0^5 = 37.5$	B1 [B1]
	For the correct area of R $\frac{2075}{12}-37\frac{1}{2}=\frac{1625}{12}=135\frac{5}{12}$	A1 [5]
	ALT	
	For writing an expression for the area of R with the correct limits. $\int_0^5 \left(15+17x+x^2-x^3\right)\,\mathrm{d}x - \int_0^5 (15-3x)\,\mathrm{d}x = \left[\int_0^5 \left(20x+x^2-x^3\right)\,\mathrm{d}x\right]$ Award the B mark for a correct expression for the combined area.	M1 B1
	For an attempt to integrate the expression for the combined area or just the curve. [Ignore limits for this mark] Area $= \int_0^5 \left(20x+x^2-x^3\right)\,\mathrm{d}x = \left[\frac{20x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\right]_0^5$	M1
	For evaluating their integral using their limits, but the lower limit must be 0. Area $= \left(\frac{20\times5^2}{2}+\frac{5^3}{3}-\frac{5^4}{4}\right)=\left(\frac{1625}{12}\right)$	M1
	For the correct final area.	A1 [5]
		Total 15 marks

Type MS · Area between two curves (or a curve and a line)

Answer 11 of 11 · Question 8 part c · 5 marks

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Question	Scheme	Marks